nucleus.

30 **A** From the first 2 sets of reading, Corrected count rate, $C_0 = 40\ s^{-1}$, $C = 10\ s^{-1}$.

$$\frac{C}{C_0} = \left(\frac{1}{2}\right)^{\frac{20}{T}}$$

$$\frac{10}{40} = \left(\frac{1}{2}\right)^{\frac{20}{T}} \Rightarrow \left(\frac{1}{2}\right)^{2} = \left(\frac{1}{2}\right)^{\frac{20}{T}}$$

By comparison, T = 10 days.